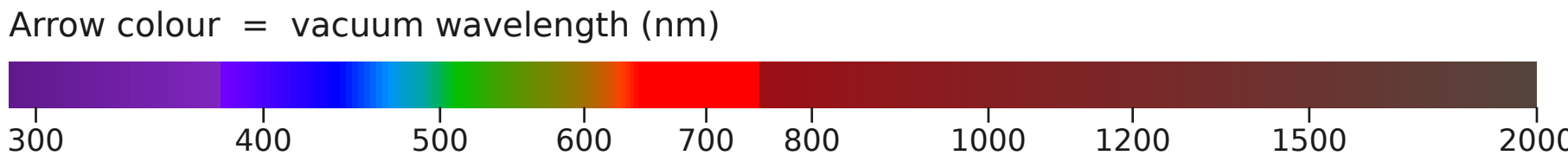


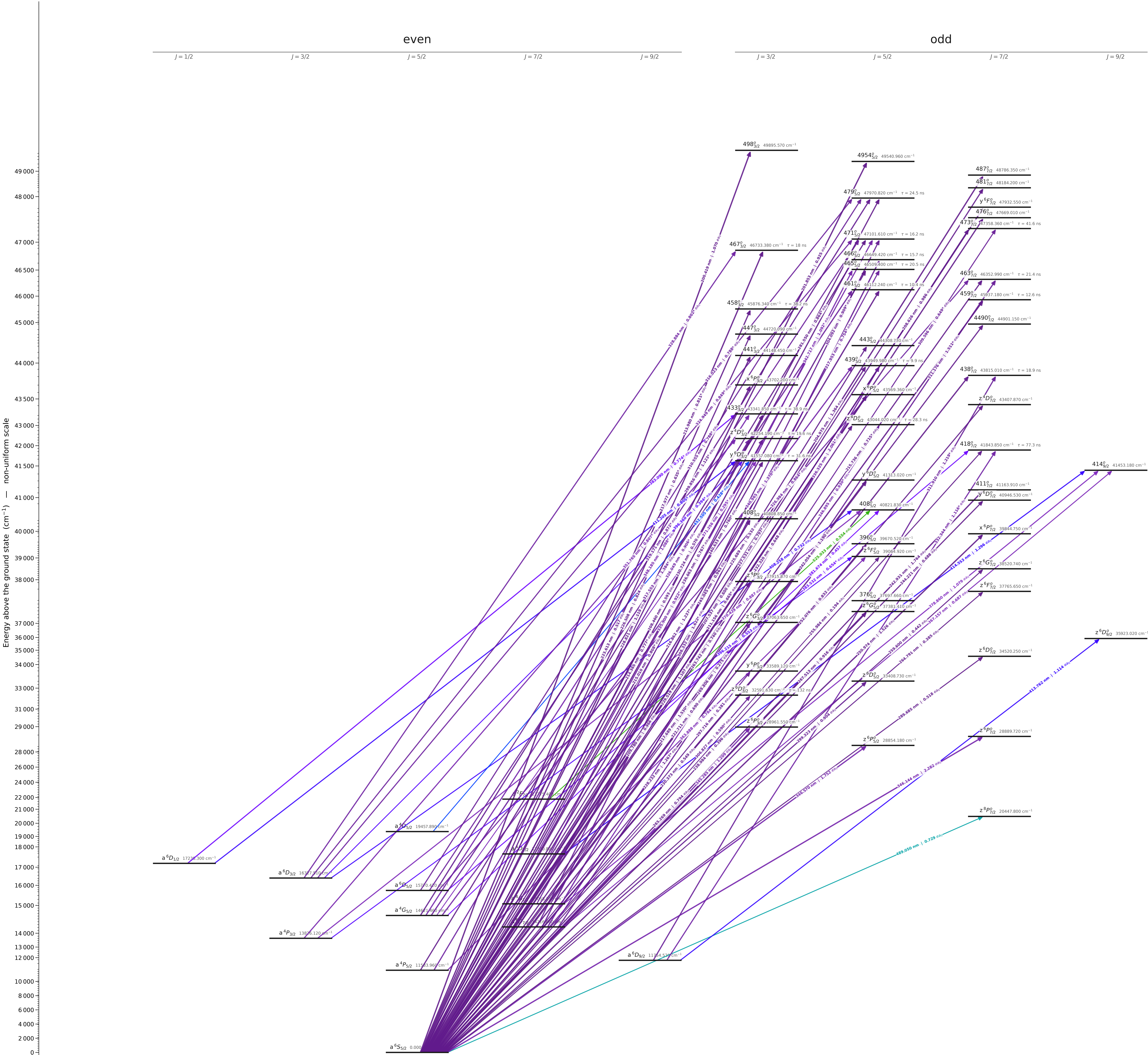
Rhenium

Arrow label: vacuum wavelength | reduced dipole matrix element $|U|er|U'|$ in ea₀
Matrix elements are measured or NIST values; * = high-accuracy theory, ≈ = model calculation.
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, $n = 70$.
Level caption: energy (NIST, or a measurement of the last 20 years where one exists) and measured lifetime (where one exists).



Re (Re I). Strongest classified transitions below 2 μm (NIST): 67 levels, 90 lines.

Re⁺ ionisation limit: 63181.60 cm⁻¹ = 7.83352 eV (158.274 nm)



Strongest transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	U er U' (ea ₀)	A (s ⁻¹)	Γ/2π up. (MHz)	branch (%)	Use
a ⁶ S _{5/2} - 498 _{5/2}	200.4186	200.3538	1495.8316	1.070	7.210e+07	-	-	-
a ⁶ S _{5/2} - 4954 _{5/2}	201.8532	201.7881	1485.2006	0.925	3.510e+07	-	-	-
a ⁶ S _{5/2} - 487 _{5/2}	204.9754	204.9097	1462.5780	1.364	5.470e+07	-	-	-
a ⁶ S _{5/2} - 467 _{5/2}	213.9798	213.9124	1401.0315	0.811*	3.400e+07	8.842	61.1	-
a ⁶ S _{5/2} - 461 _{5/2}	216.8622	216.7941	1382.4102	1.237*	5.070e+07	15.3	52.6	-
a ⁶ S _{5/2} - 459 _{7/2}	217.6886	217.6204	1377.1620	1.530*	5.750e+07	12.63	72.4	-
a ⁶ S _{5/2} - 438 _{7/2}	228.2323	228.1619	1313.5410	1.267*	3.420e+07	8.421	64.8	-
a ⁶ S _{5/2} - x ⁶ P _{5/2}	228.8214	228.7509	1310.1590	1.119	5.290e+07	-	-	-
a ⁶ S _{5/2} - x ⁶ P _{3/2}	229.5191	229.4484	1306.1765	1.107	3.420e+07	-	-	-
a ⁶ S _{5/2} - y ⁶ D _{5/2}	242.0544	241.9809	1238.5332	1.190	3.370e+07	-	-	-
a ⁶ S _{5/2} - 411 _{5/2}	242.9312	242.8575	1234.0630	1.764	5.500e+07	-	-	-
a ⁶ S _{5/2} - 408 _{3/2}	245.0449	244.9707	1223.4185	0.963	3.190e+07	-	-	-
a ⁶ S _{5/2} - x ⁶ P _{1/2}	250.9741	250.8985	1194.5156	1.526	3.730e+07	-	-	-
a ⁶ D _{7/2} - 466 _{5/2}	308.3321	308.2426	972.3037	1.622*	3.030e+07	10.14	47.6	-
a ⁶ D _{7/2} - 439 _{5/2}	336.3253	336.2287	891.3765	2.061*	3.770e+07	16.08	37.3	-
a ⁶ S _{5/2} - z ⁶ P _{7/2}	346.1439	346.0448	866.0920	2.282	3.180e+07	-	-	-

λ, ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.
Γ/2π: natural linewidth of the upper level, 1/(2πτ).